

Territorial Information Retrieval from Heterogeneous Open Data through the Construction of a Data Warehouse for Water Management in Mexico City

Eduardo Uriel Velázquez Arrieta, Omar Fernando Pulido
Morales^[0009–0001–5698–987X], Emilio García López, Clara Aide Hernández
Martínez, and Gabriel Hurtado Avilés^[0009–0002–5686–1822]

Instituto Politécnico Nacional, Mexico City, Mexico
{evelazqueza1900, opulidom2000, egarcial2001, chernandezm1903}@alumno.ipn.mx
ghurtadoa@ipn.mx

Abstract. This work presents a data warehouse for the territorial information retrieval of water consumption in Mexico City (CDMX), built from heterogeneous open sources. The contribution lies in three methodological elements: an ETL process that consolidates official SACMEX data into a unified dimensional model, a declarative R2RML mapping that republishes the resulting star schema as an RDF knowledge graph aligned with the RDF Data Cube and GeoSPARQL vocabularies, so that territorial retrieval can also be expressed as graph traversal, and a deployment methodology that guarantees an accessible demonstration through a static version published on GitHub Pages and backed by an updated public repository. The prototype organizes the information in a dimensional model and exposes it through a retrieval interface with an interactive choropleth map; filters by year, bimester, borough (*alcaldía*) and neighborhood (*colonia*); and an exploratory module for detecting atypical consumption. The analysis reveals territorial and temporal consumption patterns, turning dispersed open data into queryable information for urban diagnosis, academic research, and citizen monitoring.

Keywords: Data warehouse · Information retrieval · Open data · ETL · Knowledge graph · RDF Data Cube · GeoSPARQL · Water consumption · Mexico City · Geospatial analysis

1 Introduction

Mexico City faces a water crisis characterized by aquifer overexploitation, leaks in the distribution network, and population growth that increases demand. Although open data on water consumption exist (SACMEX), they are in poorly accessible formats and without integrated query tools, which limits their use for evidence-based decision making.

This work addresses the problem from two perspectives proper to information processing: the extraction of structured information from heterogeneous

open sources through an ETL process, and the retrieval of that information through a parameterized query interface. The purpose is to transform dispersed datasets into a queryable dimensional repository that allows retrieving, filtering, and geospatially visualizing water consumption by neighborhood and borough, so that anyone can explore it.

The contribution of this work is twofold: (i) a reproducible ETL process to consolidate official open data into a dimensional data warehouse, and (ii) a continuous-deployment methodology that keeps a functional demonstration always accessible, combining a dynamic deployment connected to a database with a static version published in a public repository.

The analysis of the 70,886 consolidated records reveals that the boroughs of Cuauhtémoc and Miguel Hidalgo lead the total consumption (Table 3), and that consumption varies across the available bimesters without a direct causal relationship with dry or rainy periods. The system is designed for five kinds of use: citizens querying consumption in their own neighbourhood, public servants diagnosing territorial demand, researchers accessing structured data for statistical analysis, journalists producing data-driven reports, and civil organisations monitoring water management.

The remainder of the article is organized as follows: Section 2 reviews related work; Section 3 describes the methodology; Section 5 presents the results and evaluation; and Section 7 sets out the conclusions and future work.

2 Related Work

2.1 Foundations: Data Warehouses, Dimensional Modeling, and ETL

Dimensional modeling and the star schema constitute the consolidated basis for the design of analytical data warehouses. Kimball and Ross [1] establish the canonical principles —fact and dimension tables, surrogate keys, slowly changing dimensions, and the ETL subsystems— that ground the model adopted in this work. On the integration phase, Vassiliadis [2] offers a reference survey of *Extract–Transform–Load* technology, covering the conceptual and logical modeling of the processes and the cross-cutting extraction, transformation, and loading issues that methodologically support our pipeline.

2.2 Data Warehouses and Information Systems for Water Management

In the water domain, the most direct antecedent is that of Soares et al. [3], who implement a data warehouse with an ETL process to monitor water supply and consumption, integrating meter readings, reservoir volumes, and geographic location from management-application databases and heterogeneous files. Duque-Méndez, Orozco-Alzate, and Vélez [4] develop a star schema with multidimensional analysis (OLAP) for historical hydro-climatological series, with emphasis on data-quality assessment. Abecker et al. [5] present a hybrid sensor-semantic

architecture (the WatERP project) that combines a relational/PostGIS store with a triple store for integrated resource management. These systems, however, mostly start from operational data, SCADA, or proprietary sensor networks, and not from open government data.

2.3 The Gap: ETL over Heterogeneous Open Data and Data Quality

Integrating open government data into dimensional warehouses poses its own challenges. Berro, Megdiche, and Teste [7] propose “content-driven” ETL processes that address precisely the semantic and structural heterogeneity, the absence of schemas, and the dispersion characteristic of open data, factors that overwhelm traditional ETL. Complementarily, Vetrò et al. [6] define a quantitative measurement framework for the quality of Open Government Data (accuracy, completeness, consistency, traceability, and freshness) and empirically report that the most frequent problems are missing metadata, incomplete data, and the absence of update information —exactly the challenges our ETL faces over the SACMEX data.

2.4 Geospatial Information Retrieval and Visualization

Purves et al. [8] offer a reference monograph on Geographic Information Retrieval (GIR) —indexing, search, and spatial ranking— that theoretically frames the territorial information-retrieval component of this work. On visualization, Han et al. [9] present an open-source web tool for exploring choropleth maps with automated class intervals and synchronous linking, a technique that directly supports our interactive choropleth visualization by borough.

2.5 Knowledge Graphs for Urban and Water Domains

Republishing a relational warehouse as RDF is a mature practice with W3C standards for both ends. The RDF Data Cube vocabulary [14] models statistical observations over conformed dimensions, which is the same structure a star schema encodes; R2RML [15] specifies the mapping declaratively, so the graph is derived rather than maintained by hand; and GeoSPARQL 1.1, whose motivations and compliance are documented by Car and Homburg [16] and benchmarked by Jovanovik et al. [17], supplies the topological predicates that a flat territorial dimension cannot express. In the water and urban domains the direction is established: Roussey et al. [18] survey knowledge-graph technologies as the next frontier for food, agriculture and water; Papageorgiou et al. manage industrial water-treatment knowledge as a graph;. What these works share is the premise this article adopts: the graph is valuable for interoperability and topological querying, not as a replacement for the analytical store.

2.6 Reproducibility and Deployment of Scientific Demonstrations

Finally, Sandve et al. [10] formulate widely cited rules for computational reproducibility (version control, documentation of preprocessing, public access to data and scripts) that ground our reproducible static version. Hill et al. [11] provide a practical guide to developing and deploying an open-source, reproducible web dashboard for water-sanitation data, a close domain that supports the continuous-deployment methodology with reproducible artifacts.

2.7 Positioning of the Contribution

Unlike existing water data warehouses—which start from operational data, SCADA, or proprietary sensors [3,4,5]— and recognizing that the integration of heterogeneous open data is a documented but unsolved problem in this domain [7,6], this work differs in three respects. First, an ETL process designed to consolidate official SACMEX open data into a dimensional star schema. Second, the exposure of the information through parameterized queries and a choropleth geospatial visualization that enables territorial information retrieval [8,9], published as static artefacts that require no server. Third, a continuous-deployment methodology accompanied by a reproducible static version that guarantees the indefinite verifiability of the system [10,11]. This same approach of consolidating official open data was previously applied to the seismic-risk domain in Mexico [12], a direct methodological antecedent transferred here to the water domain. To the best of our review, no academic, peer-reviewed, and reproducible data warehouse built specifically over SACMEX/CDMX open data exists.

3 Methodology

3.1 General Architecture

The system follows a layered architecture: a data layer (PostgreSQL), a processing layer (ETL with Python and SQL), a query layer of parameterized SQL and, over it, a declarative RDF projection, and a presentation layer (HTML/CSS/JS frontend with Leaflet and Plotly). The warehouse implements a star schema with one fact table (`fact_water_consumption`) and three dimensions: `dim_time` (year, bimester), `dim_location` (borough, neighborhood, latitude, longitude), and `dim_dev_index` (development level).

3.2 ETL Consolidation Process

The first contribution of the work is the ETL process that consolidates the SACMEX open data into the dimensional warehouse. The complete detail of the process—the SQL schema (DDL), the `loader.py` loading script, and the parameterized retrieval queries—is documented in the project’s public repository; this article presents only the architecture of the process (Fig. 1).

The pipeline operates in four phases. (1) **Extraction**: reading the open SACMEX CSV files. (2) **Validation and cleaning**: discarding records with

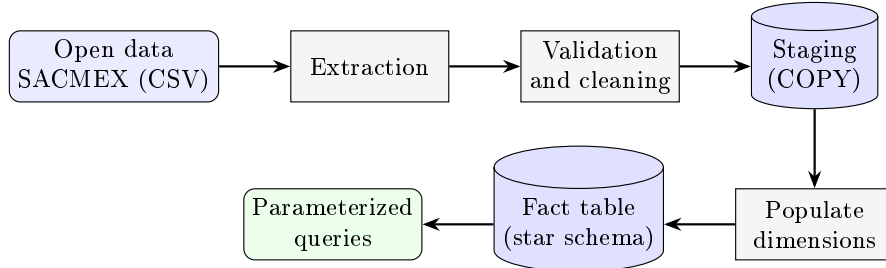


Fig. 1: ETL process architecture: from SACMEX heterogeneous open data to the star schema exposed through parameterized queries.

empty fields, inconsistent coordinates, or incompatible formats. (3) **Loading to staging and populating dimensions**: COPY of the data into temporary tables and extraction of the unique values into `dim_time`, `dim_location`, and `dim_dev_index`. (4) **Populating the fact table**: a join between staging and dimensions to populate `fact_water_consumption` under the star schema, finally dropping the staging tables. The information thus consolidated becomes available for retrieval through the parameterized queries distributed with the archived release.¹

This duality solves a recurring practical problem in academic prototypes — that the demo stops working after evaluation— and makes the work reproducible indefinitely.

3.3 Continuous Deployment Methodology

To guarantee that the demonstration remains available and reproducible without depending on the availability of a server, the system is published in two complementary modalities. The **dynamic deployment** keeps the complete architecture over PostgreSQL and serves parameterized queries and aggregations against the data warehouse; it was used to obtain the experimental results reported here, and its availability depends on the hosting service. The **static version** exports the consolidated data as JSON and GeoJSON files that the frontend consumes directly, so that querying, filtering, aggregation and visualization work on the client without an active backend. This duality addresses the recurring practical problem that academic prototypes stop working shortly after evaluation [10,11]. The system is available at:

- **Public repository**: https://github.com/gabrielhuav/Data_Warehouse_static
- **Static demo (permanent, no backend required)**: https://gabrielhuav.github.io/Data_Warehouse_static/
- **Warehouse, preservation fork**: https://github.com/gabrielhuav/data_warehouse_cdmx

¹ SACMEX operates a proprietary industrial-grade data platform (commercial technology introduced in 2021) that integrates SCADA and multiple sources; it is distinguished from a reproducible academic work over open data such as the present one.

Table 1: Records discarded during validation and cleaning, over the 71,102 records of the source export.

Exclusion criterion	Records	% of source
Unmatched territorial label	216	0.30
Missing development index	0	0.00
Reading date absent from the time dimension	0	0.00
Total discarded	216	0.30
Loaded into the fact table	70,886	99.70

What is preserved deserves precision. The archived release (https://github.com/gabrielhuav/Data_Warehouse_static/releases/tag/v1.3-icokg2026) distributes the warehouse itself —schema, ETL, both source files and the container definition that rebuilds the database, vendored unmodified from its author’s repository and additionally preserved as a fork— together with the static frontend, the R2RML mapping, a browser-side SPARQL explorer (https://gabrielhuav.github.io/Data_Warehouse_static/grafa.html) and the scripts that regenerate the exported data, so the JSON the demonstration consumes is an export of the warehouse and not a separate dataset. Citations refer to the frozen release; the repository head may evolve, the release will not.

3.4 Data Scope, Grain and Cleaning Criteria

Three properties of the source bound every result reported below. *Extent*: the resource is published as *Consumo Agua (1er Semestre 2019)* and contains exactly that —the first three bimesters of 2019, one reading date each, in 71,102 records— so the warehouse is not a multi-year series and the seasonal variation it supports is a within-semester comparison. The bound is declared by the publisher in the title of the resource, not chosen by us, and the portal reports no release after February 2023: a partial extraction published once, without versioning, which is the problem this article studies. Against the source becoming unavailable, both files the ETL consumes are redistributed with the archived release and the portal page is preserved in the Internet Archive. *Grain*: one fact row is one *manzana* (city block) for one billing period, and the export carries 22,930 distinct coordinate-bearing units over 1,553 borough–neighbourhood pairs, so the location dimension is coarser than the grain and every aggregate is a roll-up. *Scale*: the development index has four ordinal levels —ALTO, MEDIO, BAJO and POPULAR— published with the dataset rather than derived here.

The exclusion criteria are deterministic and enforced structurally: the fact table is populated by inner joins against the three dimensions, so a record that fails to resolve a dimension member is not loaded. Table 1 quantifies each criterion rather than merely declaring data quality [6]. The distribution is unusually clean and that is itself a finding: a single criterion accounts for every discarded record —the 216 rows carrying the literal NA in both territorial fields— and none is lost to a malformed measure or an unparseable date. This is a property of the SACMEX export rather than a merit of the process.

Data sources and attribution. The meteorological series comes from Open-Meteo under CC BY 4.0 and the cartographic base of the interactive map is OpenStreetMap under the Open Database License, which requires attribution wherever the map is reproduced.

3.5 Frontend and Geospatial Visualization

The frontend offers a dashboard with indicators, interactive charts (Plotly), and a choropleth map of the 16 boroughs (Leaflet + GeoJSON), where color represents the consumption level and each borough displays a popup with its detail.

3.6 Implementation of the Atypical-Consumption Module

The current version replaces the idea of a traditional supervised prediction with an atypical-consumption detection module oriented to analytical retrieval. Instead of training a model with historical anomaly labels—which are not available in the open source—the system computes statistical indicators over the filtered records and detects zones whose consumption departs significantly from the aggregate behavior.

The procedure is implemented as an unsupervised analysis on the frontend: first, the records are grouped by borough, neighborhood, year, and bimester; then reference measures such as total consumption, average consumption, and relative dispersion are computed; finally, the zones with the largest deviation are ordered to show them as possible atypical consumption. This approach avoids presenting a prediction as if it were a model trained with reference data (ground truth), but keeps the practical value of the ML/analysis component by flagging patterns that require later inspection.

Formally, for each territorial group g an atypicality score A_g is computed from the normalized deviation with respect to the mean of the filtered set:

$$A_g = \left| \frac{x_g - \mu}{\sigma + \epsilon} \right|, \quad (1)$$

where x_g represents the aggregate consumption of the group, μ is the average consumption of comparable groups, σ is the standard deviation, and ϵ prevents division by zero. The groups with the largest A_g are reported in the interface as zones with atypical consumption. Therefore, the module must be interpreted as exploratory anomaly detection and not as a causal prediction of consumption.

Two clarifications delimit what this module does. First, it is an *exploratory* detector and not a causal predictor: a high A_g states that a zone departs from its comparison set, not why. Second, it does *not* perform real-time leak detection. No leak reports, work orders or network telemetry are integrated into the warehouse, and the evaluation reported in Section 5 was obtained exclusively over synthetically injected perturbations. Its operational role is to prioritise zones for later inspection, which is a retrieval task—ranking territorial units by a computed score—and not a diagnostic one. Read that way the module is not an add-on but the ranking function of the retrieval framework: the choropleth answers “how much is consumed here”, while the module answers “which units

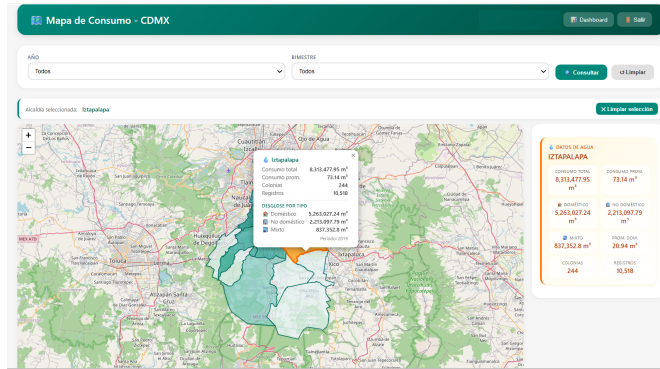


Fig. 2: Interactive map of water consumption by borough in Mexico City.

should I look at first”, which is the ordering problem at the centre of information retrieval [8].

3.7 Technologies Used

4 From the Dimensional Model to a Knowledge Graph

4.1 Rationale: Why a Star Schema, and What a Graph Adds

The star schema was not chosen by default. The dominant workload is the repeated aggregation of one numeric measure over three conformed dimensions with filters chosen interactively, which is what dimensional modelling was designed for [1]: constant join depth and aggregation by a single grouped scan. A graph would pay a traversal cost for joins that foreign keys already resolve. What a graph adds is complementary: *interoperability*, since a standard vocabulary makes the resource linkable to other municipal datasets without prior schema agreement, the recurring obstacle in the open-data literature [6]; *topological expressiveness*, since questions such as “which neighbourhoods adjacent to an atypical zone also exceed the borough mean” are natural traversals and awkward over a flat territorial dimension; and *entity reconciliation*, since dereferenceable identifiers resolve the label heterogeneity our ETL absorbs through normalisation rules. We therefore keep the star schema as the analytical store and derive the RDF layer by a declarative mapping, so that neither representation is maintained by hand.

4.2 Vocabulary Alignment and R2RML Mapping

The correspondence between a star schema and the RDF Data Cube [14] is close to mechanical, which is what makes this layer inexpensive to maintain: a fact row becomes a `qb:Observation`, the measures become `qb:MeasureProperty`, the time and location dimensions become `qb:DimensionProperty`, the neighbourhood entity becomes a `geo:Feature` with its `geo:asWKT` geometry, territorial adjacency becomes `geo:sfTouches`, and the development index becomes a

`skos:Concept` in an ordinal scheme. The mapping is expressed in R2RML [15] and distributed complete with the archived release, where it declares seven triples maps covering both fact tables, the territorial dimension and its geometry, the derived adjacency, the temporal dimension and the development-index scheme. Materialising it over the warehouse yields **761,072 triples**: 71,067 observations over 1,553 territorial features, their geometries, 181 temporal entities and the four members of the index scheme.

Two properties of the geospatial layer bound what the traversal can claim. The warehouse stores *points*, not polygons: each neighbourhood is the centroid of its readings, recovered from coordinates present in the source export but dropped by the original loading schema. The substitution is defensible at the scale at which the interface operates—the 22,930 distinct coordinates are never shared between two neighbourhoods, and the neighbourhood diameter has a median of 0.63 km—although for the ten neighbourhoods above three kilometres the point is coarse. Consequently `geo:sfTouches` expresses proximity within 1.5 km rather than shared boundaries; replacing the centroids with official polygons would make it topological in the strict sense, which is a substitution in the source data and not a change to the mapping.

4.3 Territorial Retrieval as Graph Traversal

The value of the layer is that it makes a class of questions expressible that the tabular interface does not naturally support. GeoSPARQL supplies the topological predicates [16,17]; the query below retrieves the neighbourhoods adjacent to a flagged zone together with their own consumption, a single traversal that the relational interface resolves only through an ad-hoc spatial self-join.

Listing 1.1: Territorial traversal over the knowledge graph.

```
SELECT ?neighbour (SUM(?c) AS ?consumption) WHERE {
  agua:colonia/500 geo:sfTouches ?n . ?n rdfs:label ?neighbour .
  ?o agua:location ?n ; agua:totalConsumption ?c .
} GROUP BY ?neighbour ORDER BY DESC(?consumption)
```

That the graph is queryable, and not merely declarable, is shown without any server: the static deployment ships a browser-sized subset—140,481 triples using the same vocabulary—with a SPARQL 1.1 engine running on the client, so a reader can edit a query and read the answer without installing a triple store. The two representations agree where they overlap: aggregating consumption by bimester through SPARQL, joining consumption and climate on their shared temporal dimension, returns the same totals the reproduction script obtains in SQL over the source files. The graph is a projection of the warehouse, not a second version of it.

5 Results and Evaluation

5.1 Integrated Data Volume

The ETL process consolidated the water-consumption information of Mexico City into the dimensional warehouse. Table 2 presents the stored volume. Each

Table 2: Data volume stored in the data warehouse

Table	Records
fact_water_consumption	70,886
dim_location	1,553
dim_time	181
dim_dev_index	4

cardinality answers to a different grain and this must be read carefully. The 1,553 locations are the distinct borough–neighbourhood pairs and are coarser than the fact grain. The four index levels are the four members of the SACMEX ordinal scale, not three levels plus a residual. And the 181 time records are *not* reading dates: they are the daily observations of the climate series integrated alongside the consumption data, from which the dimension is populated. The consumption source carries three reading dates only, one per bimester, so the fact table distributes its rows over those three while the climate facts use all 181. The shared temporal dimension at daily grain is what allows both to be queried together.

5.2 Findings of the Consumption Analysis

The analysis of the 70,886 consolidated records reveals the following consumption patterns in Mexico City:

- **Territorial concentration:** The boroughs with the highest total consumption are Cuauhtémoc and Miguel Hidalgo, followed by Benito Juárez and Gustavo A. Madero (Table 3).
- **Temporal distribution:** Consumption shows variations across the bimesters available in the warehouse. In the aggregated query by bimester, the highest total consumption was observed in bimester 2, followed by bimester 3 and bimester 1. Therefore, no direct causal relationship with dry or rainy periods is established without integrating additional climatic variables.
- **Relationship with the development index:** Neighborhoods with a High development index show a higher average consumption than those with a Low index. This is a descriptive observation over the warehouse; attributing it to a socioeconomic factor would require an external development-index source and is therefore left as future work rather than claimed as a causal relationship.

The consistency of the reported volumes was verified directly on the original data warehouse. The fact table contains 70,886 records, while the dimensions contain 1,553 locations, 181 time records, and 4 development-index levels. In particular, the `dim_time` dimension does not represent only year–bimester combinations, but complete dates with derived attributes (`date`, `year`, `month`, `day`, and `bimester`), so the 181 records correspond to distinct dates present in the source.

The interface dynamically generates the list of neighborhoods with the highest total consumption according to the filters selected by the user.

Table 3: Five boroughs with the highest total water consumption

Borough	Total consumption
Cuauhtémoc	18 401 765.79
Miguel Hidalgo	14 490 397.15
Benito Juárez	13 780 118.64
Gustavo A. Madero	13 480 100.57
Álvaro Obregón	9 608 545.20

Table 4: Evaluation of the atypical-consumption detection module on injected anomalies (synthetic ground truth; $N = 11,460$, 3 % anomalies, seed 42, scikit-learn 1.7). Produced by `evaluacion-ml/eval_anomalias.py` in the archived release; the figures are exactly what that script emits.

Method	Precision	Recall	F1	ROC-AUC	PR-AUC
z-score A_g (this work)	0.914	0.677	0.778	0.983	0.836
Isolation Forest	0.963	0.983	0.973	1.000	0.999
LOF	0.330	0.084	0.134	0.583	0.102

5.3 Quantitative Evaluation of the Atypical-Consumption Module

Although the module operates in an unsupervised manner in production—the open source provides no anomaly labels—its performance was evaluated through a controlled and reproducible protocol with injected anomalies acting as ground truth. Over a synthetic set with the structure of the warehouse ($N \approx 11,460$ records from 16 boroughs), 3% of labeled anomalies were injected (70% high spikes, 30% drops) and three detectors were compared: the A_g z-score of this work (threshold $A_g > 3$), Isolation Forest, and LOF, using as features the log-scaled consumption and its ratio to the neighborhood mean. Table 4 reports precision, recall, F1, and the areas under the ROC and precision–recall curves.

The z-score favours precision over recall: it flags few false positives (0.91) but recovers two thirds of the injected anomalies (0.68), detecting extreme spikes reliably while its fixed threshold misses part of the drops. The asymmetry follows from the shape of the data rather than from the statistic: consumption is bounded below by zero, so an injected drop produces a smaller absolute deviation than a spike of the same relative magnitude, and a symmetric threshold captures the spikes while leaving part of the drops below the cut. Isolation Forest offers the best trade-off ($F1 = 0.97$), while LOF proves poorly suited to this type of global-magnitude anomaly ($F1 = 0.13$), since it looks for locally sparse neighbourhoods whereas the injected perturbations are departures in global magnitude. These results support the z-score as a conservative exploratory filter, one that ranks a short list a utility can afford to inspect, and point to Isolation Forest as the natural improvement.

Three limitations of the protocol must be stated. The features used are aligned with the mechanism by which the anomalies were injected, which inflates separability and explains the near-perfect areas under the curve; the figures are an upper bound, not expected field performance. The thresholds were not

equalised across methods, so a comparison at a single operating point favours the methods whose threshold was chosen. And none of this constitutes evidence about real-time leak detection, for which the system is not designed and for which no labelled data were available.

Validity of the protocol. These metrics do not represent performance against real leaks or observed anomalies, but rather the ability of each method to recover injected synthetic anomalies under a controlled protocol. This decision is due to the fact that the open source does not provide real anomaly labels. Therefore, the results should be interpreted as a reproducible evaluation of sensitivity to known perturbations, not as a definitive operational validation.

The z-score threshold was set at $A_g > 3$, following the usual statistical criterion for extreme values. For Isolation Forest, $s > 0.60$ was used, and for LOF, $s > 1.5$, criteria defined before the comparison. In future work, the complete precision–recall curve will be reported and the thresholds will be tuned through validation with labeled data, when available.

5.4 Performance Metrics

Frontend performance was measured with Lighthouse 13.4.1 against the permanent static deployment, which is the version that remains available. On emulated desktop the first contentful paint is 0.2 s, the largest contentful paint 0.3 s, the speed index 0.6 s and the total blocking time 200 ms, for a performance score of 89; on emulated mobile over throttled 4G they are 1.1 s, 3.7 s, 1.6 s and 530 ms, scoring 76. Accessibility, best practices and search-engine optimisation score 100 on both profiles. Reaching these values required deferring the third-party plotting and mapping libraries, which were originally loaded synchronously in the document head and delayed the first paint by 2.8 s on desktop and 9.2 s on mobile. The remaining bound is structural: the static version downloads the full 3.9 MB export and filters and aggregates on the client, so what a server would compute once is computed by each visitor, and the largest contentful paint on mobile reflects that transfer. That is the price of removing the server dependency.

Query Performance over the Total Data Volume. Frontend metrics do not characterise the retrieval layer, so four analytical workloads were timed directly against the warehouse over all 70,886 fact rows: a paginated retrieval, a full-scan aggregation by borough, a full-scan aggregation by neighbourhood and bimester—the heaviest query the dashboard issues—and a top- N ranking. Each ran 200 times after ten warm-up calls on twelve logical cores with 32 GB of RAM and PostgreSQL 16.14 in a container, which is what makes the reported percentile meaningful, unlike the ten executions of a single workload reported in the submitted version. These are SQL latencies measured at the database, excluding serialisation and transport.

The profile is the expected one for a star schema of this size: the three workloads that resolve through the dimensions stay near 25 ms while scanning the full fact table, whereas the aggregation by neighbourhood and bimester costs four times more because it groups over 1,553 locations rather than sixteen. The

Table 5: Response time of the retrieval workloads over the complete warehouse (70,886 fact rows), 200 executions after ten warm-up calls.

Workload	Mean (ms)	p95 (ms)	Executions
Paginated retrieval (limit=100)	32.3	35.2	200
Full aggregation by borough	24.1	27.0	200
Aggregation by neighbourhood and bimester	101.3	125.4	200
Top-10 neighbourhoods with ordering	23.1	23.6	200

heaviest interactive query returns in about a tenth of a second, and the gap between mean and p95 stays below 25 % throughout, indicating that the cost is dominated by the scan rather than by contention.

5.5 Functional Tests

The operation of the ETL processes, the persistence in PostgreSQL, the parameterized queries and the information retrieval was verified. The tests included queries parameterized by year, bimester, borough, and neighborhood; aggregated retrieval (total and average consumption by neighborhood); generation of JSON responses; geospatial retrieval for visualization; and pagination with configurable limits. All modules produced results consistent with the expected structure.

5.6 Information-Retrieval Capability

The dimensional model allows retrieving territorial information through combined filters of time, location, and development index, executing analytical queries over 70,886 records without directly accessing the original sources. The supported operations include querying consumption by neighborhood and by borough, temporal aggregation by year and bimester, coordinate retrieval, geospatial exploration, and ordering by total consumption (top 10 neighborhoods).

6 Discussion

Against the closest antecedents the contribution is one of *provenance and preservation* rather than of scale. Soares et al. [3], Duque-Méndez et al. [4] and Abecker et al. [5] all begin from data an organisation already controls —operational readings, curated series, a sensor network. This work begins from a public export with no schema, no versioning and no documented update cadence, which moves the difficulty from modelling to consolidation and is why the cleaning criteria and their yield are reported explicitly rather than assumed. The second difference is durability: those systems depend on a running service, whereas here the interface, the graph and the data are preserved as static artefacts any reader can re-execute, addressing the recurring problem that academic prototypes stop working shortly after evaluation [10,11].

The practical implication is a caution about how the totals should be read. The boroughs that consume most are not those that consume most per inhabitant: Cuauhtémoc and Miguel Hidalgo concentrate commercial activity, and the warehouse supports the distinction because the source separates domestic from non-domestic billing —the non-domestic share reaches 32.5 % in Cuauhtémoc against 20.3 % in Gustavo A. Madero, twelve points apart between boroughs whose totals differ by barely a third. Total consumption is therefore not a per-capita indicator and should not be used as one for allocating supply interventions; the per-account average, also stored, is the appropriate measure.

7 Conclusions and Future Work

7.1 Conclusions

A data warehouse was presented for the territorial information retrieval of water consumption in Mexico City, built from open sources through a reproducible ETL process and published with a continuous-deployment methodology that guarantees a permanent functional demonstration. The solution consolidated 70,886 consumption records and 1,553 locations across the 16 boroughs into a dimensional star schema on PostgreSQL.

The findings reveal that water consumption in Mexico City is concentrated in the boroughs of Cuauhtémoc and Miguel Hidalgo (Table 3), and that it varies across the available bimesters without establishing in this work a causal relationship with dry or rainy periods. These results show that the system not only consolidates data, but also allows extracting knowledge useful for water management.

The two central contributions —the consolidation ETL and the always-accessible demo methodology (dynamic deployment + reproducible static version)— constitute a replicable method for bringing heterogeneous official open data to queryable and verifiable territorial information.

7.2 Threats to Validity and Limitations

Four limitations bound what may be concluded. The temporal extent is the first: the source publishes three bimesters of a single year, so the seasonal variation observed is a within-semester comparison and the climate correlation rests on three aggregated points, enough to describe a direction but not to estimate one. The second concerns the anomaly module, whose evaluation uses synthetic perturbations aligned with the detectors; its figures are an upper bound and say nothing about real leaks. The third is geometric: neighbourhoods are represented by centroids rather than polygons, so the adjacency projected as `geo:sfTouches` is proximity within 1.5 km and not shared boundaries. The fourth is descriptive: the association between development index and consumption is an observation over the warehouse, and attributing it to a socioeconomic mechanism would require an external source the warehouse does not integrate.

7.3 Future Work

The most immediate line is to materialise the knowledge graph into a persistent triple store with a public SPARQL endpoint, and to replace the centroid-based adjacency with official neighbourhood polygons, which would make the topological predicates exact rather than approximate. Aligning the territorial identifiers with external authorities would let the resource be joined with other municipal datasets without prior schema agreement, which is the interoperability argument that motivates the graph layer. On the retrieval side, the anomaly module should be re-evaluated when labelled leak data exist, reporting complete precision–recall curves instead of a single operating point, with Isolation Forest as the candidate production detector. Finally, the method transfers to other open urban-data domains —mobility, air quality and seismic risk— where the same combination of dispersed official files and absent query tooling recurs.

Declaration on the Use of Generative AI

The authors used a generative AI assistant under their direction during the preparation of this camera-ready version. Its scope was language editing and L^AT_EX formatting; drafting passages the authors then revised and approved, principally the statements of scope, provenance and attribution added for the reviewers; writing the auxiliary scripts distributed with the artefact, the R2RML mapping and the SQL restoring the coordinates; and cross-checking the manuscript against the published data, which is how several inconsistencies with an earlier draft were located. The research questions, the dimensional design, the ETL, the data collection, the experiments and the interpretation are the authors’ own work. No figure reported here originates with the assistant: every quantity was recomputed from the two published source files and independently checked, and the scripts that perform that recomputation are distributed with the release. The authors verified every cited reference against its original source, confirm that no text was reproduced from third-party works, and take full responsibility for the content.

Disclosure of Interests

The authors declare that they have no conflicts of interest relevant to the content of this article.

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